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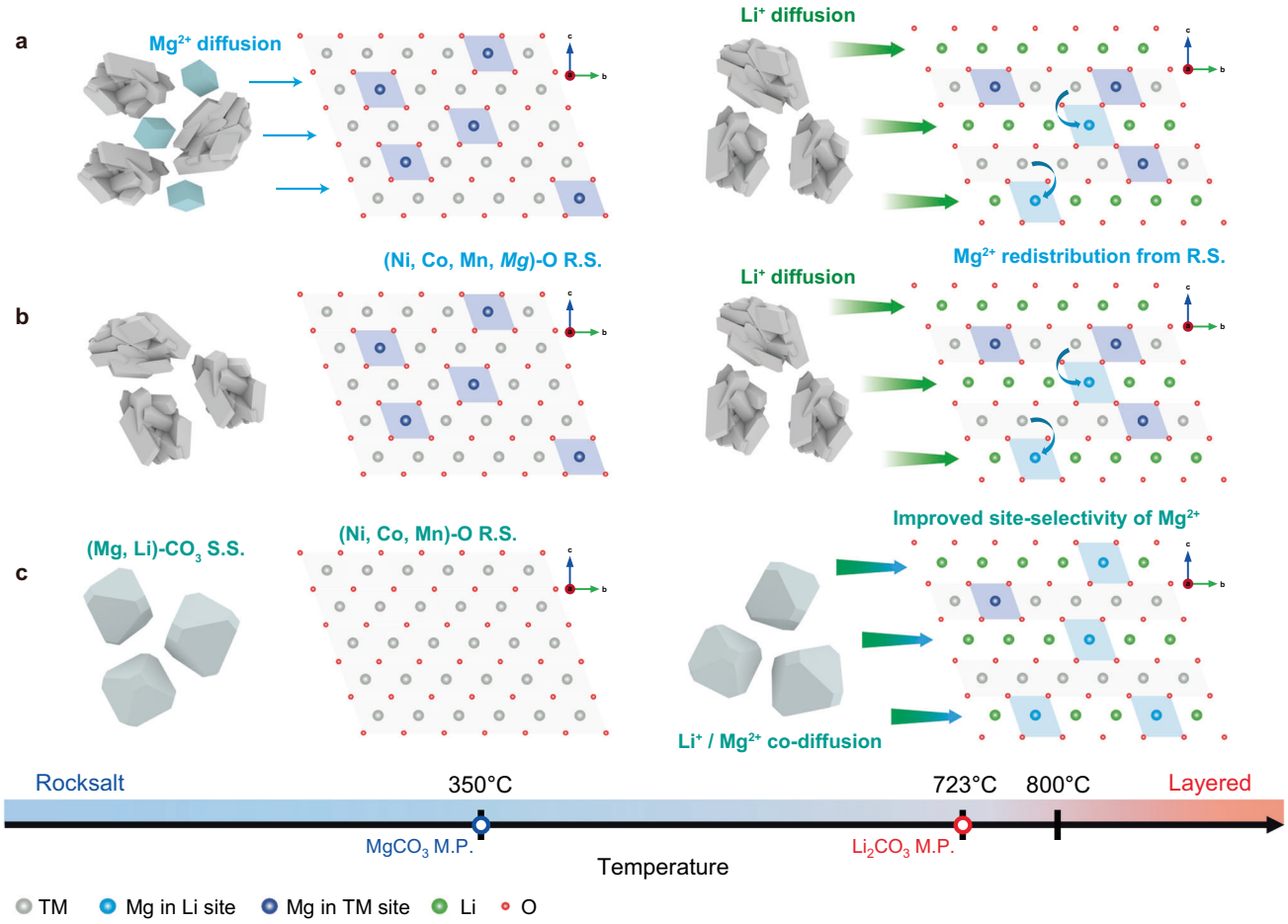

Fig. 5 | Kinetically controlled site selectivity of Mg^{2+} by adjusting the incorporation timing. a–c Schematic diagrams of the thermal diffusion pathways of Li^+ and Mg^{2+} during cathode materials synthesis are shown. The TM hydroxide precursor turned to a rocksalt structure as the temperature increased. SSD (**a**) and CPD (**b**) first constructed a (Ni, Co, Mn, Mg)–O rocksalt phase, and Mg^{2+} were redistributed inside the layered structure as Li^+ started to diffuse. **c** However, in the case of LCD, Li^+ and Mg^{2+} co-diffused to the (Ni, Co, Mn)–O rocksalt structure, which improved the site selectivity of Mg^{2+} to the Li site.

188.3 and 246.1 USD (LC ton)$^{-1}$, respectively, which respectively correspond to 5.9% and 3.5% reduction from the total OPEX (Table 1). All consumables, reagents, utilities, and waste disposal costs were partially reduced with the decreased process load (see the details in Supplementary Tables S18 and S19). The major cut in cost was attributed to the less use of Mg elimination agents, which not only reduced reagent cost, but also reduced by-product leading to utility and waste disposal cost reduction. Moreover, from the P/P scale Mg refinery test using brine from Argentina (Supplementary Tables S20 and S21), 79.9 USD of reagent cost was diminished per 1 ton of Li_2CO_3 production.

Table 1 | Economic and environmental evaluation of overall processes

LX process		CAPEX[a] (M $) [%]	OPEX[b] ($/LC ton) [%]	CO_2 emission[c] (eq. ton/LC ton) [%]
Brine[d]	As is	61.8	3221	9.47
	Reduced	49.8 [−19.4%]	3032.7 [−5.9%]	8.615 [−9.0%]
Hard rock[e]	As is	133.6	6995	7.45
	Reduced	124.1 [−7.3%]	6748.9 [−3.5%]	7.125 [−4.4%]

[a]Reduction in equipment cost by millions of USD and reduction ratio from the total equipment costs (Supplementary Tables S15 and S16).
[b]Reduction in OPEX in USD per 1 ton of LC production and reduction ratio from the total OPEX (Supplementary Tables S18 and S19).
[c]Reduction in CO_2 emission by equivalent ton per 1 ton of LC production and reduction ratio from the total CO_2 emission (Supplementary Tables S22 and S23).
[d]Brine LX process (Supplementary Fig. S1).
[e]Hard rock LX process (Supplementary Fig. S2).

The environmental impact was evaluated with respect to CO_2 emission. As environmental, social, and governance pressures are ever-increasing worldwide, the reduction of the carbon footprint of the lithium production sector is highly critical for sustainable LIBs. Supplementary Tables S22 and S23 compare the carbon footprints of the conventional and less-purifying Mg LX processes for brine and hard rock, respectively. The Mg refinery reagent $Ca(OH)_2$ has a high carbon footprint because quicklime (CaO), the precursor of $Ca(OH)_2$, requires the high-temperature calcination of limestone and produces CO_2 as a product. Thus, the reduction of the usage of this reagent could curtail CO_2 emissions. Furthermore, reductions in the energy and waste carbon footprints were achieved by the IX process and $Mg(OH)_2$ cake reduction (see Supplementary Tables S22 and S23). Overall, the CO_2 emissions in the brine and hard rock LX processes were reduced by 9.0% and 4.4% of the total CO_2 emissions, respectively (Table 1), achieving a more sustainable LX process. However, the potential benefits from the enhancements in battery performance and filtration efficiency by Li_2CO_3 particle morphological effects were not considered in this analysis. Thus, additional reductions in production costs and carbon footprints are expected.

Discussion

From salt lakes to batteries, the LIBs used today are undergoing intense scrutiny from the perspectives of sustainability and the pursuit

of higher energy density. In this work, we comprehensively re-evaluated the electrochemical, economic, and environmental impacts of the battery-grade purity of Li sources. We discovered that Mg impurities of about 1% have unique benefits in total production cost and CO_2 emission during LX processes. Moreover, the inclusion of Mg^{2+} ions in the Li source resulted in the thermal diffusion of the dopant at a higher temperature, which reinforced the pillar effect and resulted in electrochemical advantages. Whereas many studies aimed to reduce the costs of TMs by controlling redox chemistry, we addressed the general belief on the battery-grade purity of Li sources and concluded that precursors with slight amounts of impurities can maximize the productivity and eco-friendliness of batteries. Hence, we suggest a reconsideration of the meaning behind industrial and commercial specifications on battery-grade purity to accomplish ultimate material stability for LIBs.

Methods

Materials synthesis

Lithium carbonate powder was synthesized via the precipitation method. Aqueous solutions with 10 g L^{-1} of Li^+ and different amounts of Mg^{2+} (0.1, 0.3, 0.5, 0.7, and 0.9 g L^{-1}) ions were prepared by dissolving $Li_2SO_4{\cdot}H_2O$ and $MgSO_4$ with appropriate ratio, and sodium carbonate saturated solution was prepared as a precipitation agent. 90 ml of a saturated sodium carbonate solution was added into a 0.5 L batch reactor containing 300 mL of lithium and magnesium sulfate solution under a feeding rate of 3 mL min^{-1}. A reaction was conducted at 90 °C for 1 h. Precipitated Li_2CO_3 powder was gathered by filtration and then washed by boiling deionized water. The washed powders were dried and carefully ground in a mortar.

$Ni_{0.6}Co_{0.2}Mn_{0.2}(OH)_2$ hydroxide precursors (Bare precursor) were prepared via a typical coprecipitation method. The 1 L of 1.0 M NH_4OH solution was purged under N_2 before reaction to remove other residual reactive gas species. 2.0 M aqueous solution of mixed $NiSO_4{\cdot}6H_2O$, $CoSO_4{\cdot}7H_2O$, and $MnSO_4{\cdot}H_2O$ with a Ni:Co:Mn molar ratio of 6:2:2 was pumped into the reactor under N_2 atmosphere. For Mg 1.5% and 3.5% precursor, certain amounts of $MgSO_4$ were added into the mixture solution. Concurrently, 5.0 M of NaOH solution and the desired amounts of NH_4OH solution as a chelating agent were separately pumped into the reactor. The pH value in the reactor was maintained 10.5 at 50 °C with a stirring speed of 1500 r.p.m. with an overhead stirrer. The obtained hydroxide precursors were filtered, washed with deionized water, and dried in a vacuum oven overnight.

Control $LiNi_{0.6}Co_{0.2}Mn_{0.2}O_2$ was prepared by thoroughly mixing the Bare precursor with a stoichiometric amount of LCB. In cases of LCD and LCD-HC, prepared LCM and LCM-HC were mixed with Bare precursor, respectively. Likewise, SSD and SSD-HC powder were prepared with stoichiometric amounts of $MgCO_3$ hydrate with a Bare precursor and LCB mixture. For CPD and CPD-HC, Mg 1.5 and 3.5% precursor were mixed with LCB instead of Bare precursor, respectively. Li/TM molar ratios of all mixtures were fixed with a 1.05 ratio to compensate for lithium loss under high-temperature conditions. All mixtures were calcined at 850 °C for 12 h in a tube furnace under O_2 gas flowing.

For better understanding, a visual summary of synthesized materials is provided in Supplementary Note S1.

Electrochemical testing

The obtained active materials were mixed with a carbon black and polyvinylidene fluoride (PVDF) in a weight ratio of 90:5:5 (active material:carbon black:PVDF) in *N*-methyl-2-pyrrolidone (NMP). The slurry was cast onto the Al foil current collector (5–6 mg cm^{-2} loading of active materials). The electrochemical performance of half cells was evaluated using 2032 coin cells with a Li metal (counter/reference electrode) and 1 M $LiPF_6$ in EC:DEC (50:50 vol%, electrolyte). The Li metal electrode was prepared by punching the Li metal foil into a disc shape. The half cells were then cycled between 3.0 and 4.3 V (vs. Li^+/Li) using WBCS3000 (WonATech) in a temperature-controlled chamber. Every cell was rested over 10 h before cycling and pre-cycled twice.

The electrochemical performances of full cells were evaluated using 2032 coin cells with a graphite anode and 1.3 M $LiPF_6$ in EC:DEC (30:70 vol%, electrolyte). The cathode and anode (graphite anode powder, Sigma-Aldrich) active materials were loaded 13–14 and 7–8 mg cm^{-2}, respectively onto the Al and Cu foil current collector, respectively. The graphite anode was mixed with a carbon black, styrene-butadiene rubber (SBR), and carboxymethyl cellulose (CMC) in a weight ratio of 94.5:1.3:2.8:1.4 (active material:carbon black:SBR:CMC) in deionized water. The N/P ratio and the overhang fraction were controlled to ~1.1 and 12.9%, respectively. After cell assembly, the full cells were wetted at 1.5 V for 12 h and formation cycle was conducted between 3.0 and 4.2 V (vs. graphite) using WBCS3000 (WonATech) in a temperature-controlled chamber at 40 °C. The optimization of full cells was based on the recent report by Garayt et al.[36].

XRD and Rietveld refinement

XRD spectra of lithium carbonates were obtained on the PLS-II XRS-GIST 5D beamline at the Pohang Accelerator Laboratory (PAL) with a wavelength of 1.24007 Å. The XRD (D8 ADVANCES, Bruker) spectra of precursors and cathode materials were measured with a Cu Kα wavelength of 1.5418 Å. Rietveld refinements of the collected XRD patterns were carried out using FullProf Suite package[37].

Neutron diffraction and Rietveld refinement

ND data of $LiNi_{0.6}Co_{0.2}Mn_{0.2}O_2$ was obtained from HANARO facility at the Korea Atomic Energy Research Institute (KAERI). The measurement was conducted in the 2θ range of 10–160° with a step size of 0.05° using a constant wavelength of 1.834583 Å. Rietveld refinements quantitative atomic analysis of the collected ND patterns were also carried out using FullProf Suite package[37].

Neutron diffraction pattern simulation

ND pattern simulation was conducted by Pattern calculation (Neutron −CW) in FullProf Suite package[37]. The simulated patterns depended on Mg doping site were acquired by adjusting occupancy parameters based on Rietveld parameters of pure $LiNi_{0.6}Co_{0.2}Mn_{0.2}O_2$.

X-ray absorption

XAS spectra of cathode materials were obtained on the PLS-II 10 C Wide XAFS beamline at the Pohang Accelerator Laboratory (PAL). All powder samples were analysed between Kapton tape with appropriate thickness for transmission mode detection. Spectra fitting was conducted by ATHENA software package[38] and all spectra were aligned by reference foil.

X-ray photoelectron spectroscopy

XPS spectra of lithium carbonates were obtained on the PLS-II 10A2 HR-PES II beamline at the Pohang Accelerator Laboratory (PAL). Mg 2*p* XPS spectra were obtained at 961.06 eV with a pass energy of 50 eV. The spectra were obtained under ultra-high vacuum (~10^{-9} Torr) after aligning Au 4*f* reference peak position.

pH meter

pH meter (913 pH Meter, Metrohm) was used to trace the pH value in the lithium carbonate synthesis reactor. pH meter (916 Ti-Touch, Metrohm) and dispenser (800 Dosino, Metrohm) were used to maintain the pH value in the hydroxide precursor synthesis reactor. The pH meters were calibrated before measurement by three standard solutions.

ICP-OES

The chemical compositions of the lithium carbonates, hydroxide precursors, and cathode materials were determined by an ICP-OES (PerkinElmer Optima 5300 DV, PerkinElmer, Inc.). Lithium carbonates

were dissolved in an HCl solution. Hydroxide precursors and cathode materials were dissolved in *aqua regia* ($HCl:HNO_3 = 3:1$). All dissolved samples were diluted in deionized water. Calibration curves were generated using at least three standard solutions, with the results used only from correlation coefficients that were greater than 0.999 and relative standard deviation (RSD) <5%.

Scanning electron microscopy
High-resolution SEM-EDS images were obtained by FE-SEM (JSM 7800 F PRIME with Dual EDS, JEOL) and EDS (Aztec, Oxford) with an accelerating voltage of 15 kV. SEM analyses of hydroxide precursors and cathode materials were carried out on a TE-SEM (Genesis-1000, EMCRAFT) with an accelerating voltage of 10 kV. Cross-section SEM-EDS images were obtained by FE-SEM (JSM 7100F, JEOL) and EDS (Aztec, Oxford) with an accelerating voltage of 15 kV. The Li_2CO_3 powder sample was casted on Al foil with PVDF binder and polished by a cross-section polisher (SM-09010, JEOL).

Intermediated mixture analysis
The intermediated mixture analysis was conducted. The intermediated mixture sample of LCD was prepared by mixing Bare precursor with LCM lithium precursor. Likewise, the intermediated mixture sample of SSD was prepared by mixing Bare precursor with LCB lithium precursor with a stoichiometric amount of $MgCO_3$. All the mixtures were then calcinated at 400 °C for 5 h in a tube furnace under O_2 gas flowing. The chemical composition analysis was conducted by SEM-EDS. SEM backscattered electron images and EDS line scan images were obtained by FE-SEM (JSM-7100F) and EDS (Aztec, Oxford) with an accelerating voltage of 15 kV. SEM-EDS mapping images were obtained by FE-SEM (JSM 7800 F PRIME with Dual EDS, JEOL) and EDS (Aztec, Oxford) with an accelerating voltage of 15 kV. The crystal structure analysis was conducted by XRD (D8 ADVANCES, Bruker) with a Cu Kα wavelength of 1.5418 Å.

Lithium carbonate clogging test
Lithium carbonate clogging test was conducted in 2 L batch reactor with overhead stirrer. 1.2 L of aqueous solution was prepared in the reactor with a concentration of 10 g L^{-1} of Li^+. In the case of the reaction with residual Mg impurity, 0.3 g L^{-1} of Mg^{2+} was added. Both tests were conducted in the same reactor and stirrer. Sodium carbonate saturated solution was pumped into the reactor under a feeding rate of 12 ml min^{-1} for 30 min. After that, an additional reaction for 30 min was conducted to meet equilibrium maintaining 90 °C and a stirring speed of 400 r.p.m. with an overhead stirrer. The reacted batch was rested for a certain period to settle Li_2CO_3 powder down. At the same resting time, the status of both reactor batches was observed.

Pilot plant scale lithium carbonate extraction test
Through the pilot plant scale (annual plant production capacity 2500 tons of Li_2CO_3 per year) lithium carbonate manufacturing test, the behaviour of lithium and impurities in each stage were tested. Brine LX P/P test was conducted using actual brine pumped from 'Hombre Muerto' Salt Lake in the Argentine highlands and the hard rock LX P/P test was conducted using spodumene mined from Western Australia and concentrated through the following pretreatment processes. The concentrations of all samples were analysed by ICP–OES (SEPCTRO ARCOS, AMETEK, Inc.). In particular, in the case of the Mg refinery test, the industrial grade of $Ca(OH)_2$ (85% purity) was used as a reagent reflecting the conditions of the actual manufacturing process.

Particle size distribution analysis
Ex-situ particle size distribution (PSD) analysis of LCM and LCB was carried out using a particle size analyser (Mastersizer 3000, Malvern). The synthetic conditions for LCM and LCB were identical to those described in the "Material synthesis" section, with the only variation being the reaction time. The PSD analysis result of $Mg(OH)_2$ byproduct was obtained by particle size analyser (Mastersizer 3000, Malvern).

Economic and environmental analysis
The results of economic evaluation and CO_2 emission presented in this work are based on the results of actual brine and hard rock LX processes (annual plant production capacity 25,000 tons of Li_2CO_3 per year). Raw materials and reagents input during the process were calculated by applying consumption data from the actual plant. In addition, the conditions of the LX process reflect the operating conditions in the Argentine Salt Lake. The processing cost and economic feasibility were evaluated using the mass balance and energy balance derived from the actual plant. The calculation of CO_2 emission is based on this result as well.

Data availability
All data analysed and generated during this study are included in the article and its Supplementary Information. Source data are provided with this paper.

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